Water Research 287 (2025) 124409

Contents lists available at ScienceDirect

Water Research

journal homepage: www.elsevier.com/locate/watres

Removal, transport, and transformation of organic micropollutants in managed aquifer recharge: Insights from target and non-target analyses

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ARTICLE INFO

Keywords:
Micropollutant biodegradation
Water reuse
Drinking water quality
Transformation products
Non-target screening

ABSTRACT

Managed aquifer recharge (MAR) systems can reduce the stress on groundwater resources by intentionally infiltrating and abstracting (surface) water for drinking water production. Organic micropollutant (OMP) removal and transformation products (TPs) formation in MAR depend on several factors, including their sorption and biodegradation potential. Via target and non-target analyses, we monitored OMPs (parent compounds + TPs) from infiltrated water to abstracted groundwater and drinking water. Our results show that MAR improves water quality, with 61 % of the 31 OMPs in the infiltrated water being removed by >90 %. However, persistent contaminants, including artificial sweeteners and PFAS, were not removed and reached drinking water. The charge state of hydrophilic OMPs determined transport, which was driven by electrostatic interactions with soil: positively charged OMPs were retained near the infiltration point, while negatively charged OMPs travelled to the groundwater abstraction wells. Our results reveal important blind spots in the standard list of priority OMPs measured to assess drinking water quality. Specifically, non-target analysis revealed 50 % more OMPs than those detected using target analysis, and uncovered many overlooked TPs. 46 TPs were detected in groundwater but were absent in infiltrated water, indicating possible OMP biodegradation in MAR. However, 28 possibly persistent TPs were also detected in the MAR system. Therefore, post-treatment technologies targeting persistent and mobile contaminants should be applied to complement MAR at the drinking water treatment plant. Our findings can contribute to optimizing MAR systems for improved OMP removal and highlight the need for a more comprehensive approach to water quality assessment for cleaner drinking water production.

1. Introduction

In the Netherlands, freshwater scarcity due to intensive water use and droughts has increased the interest in water reuse technologies to safeguard groundwater resources. Managed aquifer recharge (MAR) is a nature-based technology employing intentional infiltration or injection of recycled water, including surface water and treated wastewater, into groundwater (Dillon et al., 2009). MAR has been in use for decades in many countries, mainly for drinking water production, as it ensures a stable water quantity for stressed groundwater systems (Dillon et al., 2019; Maeng et al., 2011; Mumberg et al., 2024). However, concerns about its effect on natural groundwater quality remain, due to the presence of organic micropollutants (OMPs), including polar and persistent compounds, in the infiltrated water (Drewes, 2009; Maeng et al., 2011). MAR systems are not specifically designed for OMP removal, causing OMPs to potentially accumulate, leach, or transform in the subsurface, thus possibly impacting drinking water sources.

Infiltration in MAR can be achieved via ponds that allow the water to percolate through soil, where chemical, physical, and biological processes alter water quality (Regnery et al., 2017). Once in the subsurface, the OMPs infiltrating with water are removed primarily via biodegradation, but they can also undergo sorption or other abiotic transformation processes (Narain-Ford et al., 2020). However, OMP transformation is often incomplete and leads to the formation of transformation products (TPs), which are generally more mobile and may be more persistent or toxic than the parent compounds (PCs) (Hübner et al., 2022; Oberleitner et al., 2021).

Knowledge of OMP fate, TP formation, and their transport in MAR

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https://doi.org/10.1016/j.watres.2025.124409
Received 29 April 2025; Received in revised form 12 August 2025; Accepted 13 August 2025
Available online 14 August 2025
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systems is limited (Mumberg et al., 2024; Specker et al., 2025). TPs are often overlooked because they are challenging to detect and partly unknown. Nonetheless, there is increasing focus on their identification for a more comprehensive environmental chemical risk assessment, especially using high-resolution mass-spectrometry (HRMS) technologies, which allow broader OMP detection (Escher and Fenner, 2011; Ore et al., 2025; Zahn et al., 2024). OMP fate and TP formation in the subsurface depend on several factors, including OMP physicochemical and intrinsic properties (e.g., charge state, mobility, persistence) and in situ environmental conditions, including hydrogeology, redox conditions, dissolved organic carbon availability, soil and microbial compositions (Maeng et al., 2011; Zhou et al., 2025). Additionally, Dutch MAR system conditions differ from other countries due to the generally high groundwater tables in the Netherlands. The unsaturated zone between the ground surface and the groundwater table is thin, resulting in more anaerobic conditions and allowing the artificially infiltrated water to mix with naturally infiltrating groundwater almost immediately. These factors must be considered to understand OMP fate and TP formation in MAR.

Although TPs are not specifically listed in current European water regulations, they are indirectly addressed. The Water Framework Directive (WFD) sets groundwater quality standards for pesticides and their relevant metabolites, and the proposed *Amendment to the WFD, the Groundwater Directive, and the Environmental Quality Standards Directive* extends these standards to include active pharmaceutical substances and relevant pharmaceutical TPs (Directorate-General for Environment, 2022a; European Parliament, Council of the European Union, 2000, 2006). These regulations provide general criteria for assessing TP relevance, focusing on toxicity, persistence, and potential bioaccumulation, yet they often leave room for interpretation. Given the uncertain but increasing inclusion of TPs in regulation, studying them is important to assess their environmental presence, potential exposure, and relevance for regulatory inclusion.

Our study investigated the occurrence of OMPs (including PCs and TPs) in a Dutch MAR system with infiltration ponds, which is used to produce drinking water and to compensate for groundwater abstraction. Through liquid chromatography (LC)-HRMS-based target and non-target analyses of water samples collected along the water infiltration path, we aimed to: (1) assess the extent to which MAR can improve water quality for drinking water production, taking into account both target and non-target analyses of OMPs; and (2) infer the effect of OMP charge state, mobility and persistence on their fate and removal in MAR. We present novel insights into the occurrence and behavior of (unknown) OMPs in MAR systems, contributing to a more comprehensive exposure assessment and improved management of OMPs in the environment.

2. Materials and methods

2.1. Description of the site

The samples were collected from a drinking water production area in the east of the Netherlands, managed by Vitens drinking water company. The soil in the area is mainly composed of sand and clay layers. The study area consists of multiple groundwater abstraction fields that supply drinking water to nearby municipalities. Our study focuses on the Olden-Eibergen abstraction field, where groundwater has been abstracted since 1935 through six wells at 19–30 m depths below ground level (phreatic abstraction).

The groundwater in the area is supplied by natural infiltration of rain and surface water and by artificial active infiltration via a MAR system since 1975. Water is diverted from the river Berkel to infiltration ponds, which account for approximately 20 % of the abstracted water at Olden-Eibergen. River Berkel's water quality is affected by five wastewater treatment plants (WWTPs) in Germany and agricultural runoff along the waterway. The land use in the groundwater catchment and protection area is mainly forest and agricultural, with grassland and arable farming that may contribute to pesticides or veterinary pharmaceuticals leaching.

The raw water from Olden-Eibergen is further treated in the nearby drinking water treatment plant, mainly by sand filtration and softening (details in SI 2.1). The water is mixed with water from other abstraction fields and production stations and stored in a subsurface reservoir before distribution.

Before this study, a bromide tracer test estimated that in Olden-Eibergen the infiltrated water travels through the subsurface for approximately one year before reaching the abstraction wells (Cirkel, 2017). Bromide breakthrough was followed from the injection point near the pond to the abstraction well (AW) with the highest infiltration contribution (AW1). Our study includes data from two tracer test monitoring wells (M1 and M2), AW1, and another abstraction well (AW2) within the infiltration area but not included in the tracer test (Fig. 1).

2.2. Sample collection

A sampling campaign at 11 sampling points was performed in June 2023 (Figure S1, Table S1), including groundwater from two depths of the monitoring well I (proxy for the infiltrated water composition), three depths for each of the monitoring wells M1 and M2 and abstracted water from two AWs (AW1 and AW2) within the infiltration area. A drinking water sample was taken at the production facility, composed of a mixture of abstracted groundwater from the Olden-Eibergen and other nearby abstraction fields (Table S2). These groundwater locations were selected based on their position along the water infiltration path through the subsurface. The details of the water sampling procedure are reported in SI 2.2.

Macrochemistry data and OMP concentrations over 5 years (2015–2020) were shared by Vitens and used as supporting data for our in-depth analysis of a single time point.

2.3. Environmental data collection

During the sampling campaign, pH, electroconductivity, and temperature were measured in situ, and water samples were taken for chloride, nitrate, nitrite, manganese, iron, sulfate, bromide, carbonate, carbonic acid, methane, and dissolved organic carbon measurements. Vitens performed these analyses and provided the results (relevant measured parameters are reported in Table S3).

2.4. OMP analysis

The samples were analyzed with target (both by Vitens and at the University of Amsterdam - UvA) and non-target analyses (at UvA) to quantify and qualify OMPs, respectively. Vitens' target data were mainly used to discuss results, whereas UvA target data were used as a support for non-target data analysis only (e.g., OMP identification level assignment).

Vitens' target analysis used high-pressure liquid chromatography (HPLC) coupled with Triple Quad/Trap mass-spectrophotometry to quantify OMPs in samples via direct injection, using a target list of 371 compounds. SI 2.4 and Table S4 provide more information on Vitens' instrumentation and method.

Before UvA target and non-target analyses, samples were concentrated by solid phase extraction (SPE) with Oasis HLB (200 mg) from Waters™, following the method as previously described (Albergamo et al., 2018). Before SPE, the samples, procedural blanks (i.e., Milli-Q water), and controls (i.e., Milli-Q spiked with 98 target OMPs at 125 ng/L) (40 mL) were spiked with a mixture of 36 isotope-labeled internal standards at 50 ng/L. After cartridge loading, washing, elution with methanol, and evaporation under nitrogen to 0.1 mL, the extracts were diluted 1:5 in MilliQ water, resulting in a final labeled internal standard

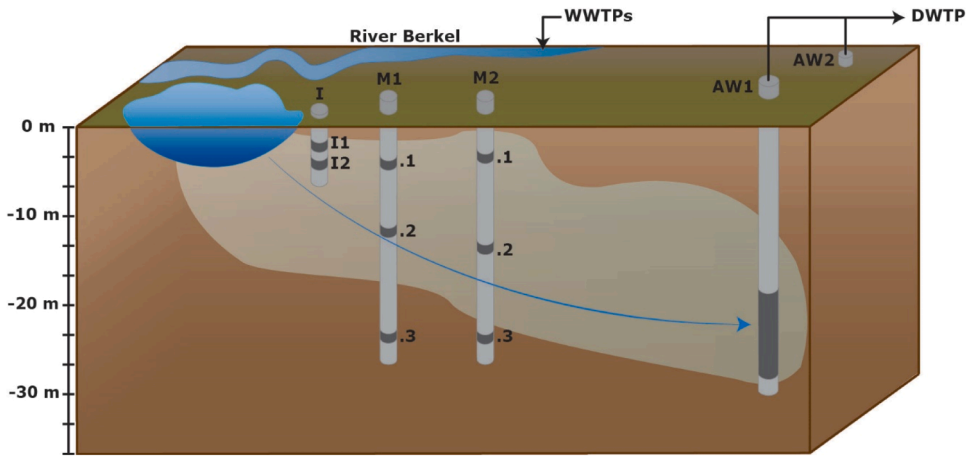


Fig. 1. Schematic representation of the MAR system. Water from the river Berkel (receiving wastewater treatment plant -WWTP- effluents) is infiltrated via a pond. The monitoring wells (I, M1, and M2) are along the infiltration path to the abstraction well that attracts the most infiltrated water (AW1) from the pond. The abstracted water from AW1 and AW2 is pumped to the drinking water treatment plant (DWTP). The sampling points' depth refers to the scale on the left of the figure.

concentration of 4 μg/L and an enrichment factor of 80. The extracts were stored at −20 °C until analysis with target and non-target methods using a Bruker Daltonics system composed of ultra-high-performance liquid chromatography (using a Kinetex® UPLC Biphenyl column) coupled with trapped ion mobility spectrometry, quadrupole time of flight, and high-resolution tandem mass spectrometry (SI 2.4 and Tables S5 and S6 report more information on UvA target and non-target methods).

2.4.1. Non-target analysis (NTA) data treatment with patRoon

The non-target data treatment was performed through open-source software, including TIMSCONVERT 2.0.0 (Luu et al., 2022) for converting raw .tdf data to .mzML format, and patRoon 2.3.3 (Helmus, 2024) for screening TPs and predicting metabolic pathways. During conversion with TIMSCONVERT, the raw data was centroided and the trapped ion mobility arrays were excluded to allow compatibility with patRoon. The script is similar to our previous study (Ore et al., 2025), but algorithm-specific parameters were optimized for the instrumentation and dataset (Supplementary Script). Specifically, OpenMS (Röst et al., 2016) feature finding and filtering parameters were optimized by adjusting *chromFWHM* (peak width at half the peak height), *chromSNR* (signal-to-noise ratio), and the absolute minimum peak intensity. After subtracting blanks (instrumental for the controls and SPE procedural for SPE samples), only peaks detected in all triplicate measurements were retained. A first suspect screening was performed to screen for 158 OMPs. The compounds detected in our samples with target analysis by Vitens, the spiked target OMPs in the SPE controls, and the spiked isotope-labeled internal standards were included in this suspect list. MzMine (Schmid et al., 2023) was used to inspect chromatograms for non-detected OMPs expected to be found in the samples based on target analysis results and spiked OMPs. Mostly, non-detection was due to peak intensity below the threshold applied in the script or presence of the same peak in the blanks (e.g., trifluoroacetic acid – TFA). A second suspect screening was then applied to screen for TPs up to the third generation that may have been formed from 410 OMPs, which included compounds regularly targeted by Vitens ($n = 276$) and those spiked in the SPE control samples ($n = 134$). The TP suspect list was generated via BioTransformer (Djoumbou-Feunang et al., 2019) with the environmental microbial degradation module (enviPath – (Wicker et al., 2016)), and included >15,000 possible TPs (Supplementary Excel, Table E2). In addition to the prioritization steps previously applied (Ore et al., 2025), when multiple suspects with the same identification (ID) level were assigned to the same feature group, the priority was given first to PCs among the suspects (detected in the control samples or target analysis), then to the suspect with the highest compound candidate ranking. Feature groups assigned exclusively to TP candidates were removed if detected in control samples, as these should not contain biodegradation TPs, suggesting potential misannotation. Also, TPs formed later than the third stage of the transformation pathway were not addressed. Additionally, every OMP and potential TP thereof with a peak intensity lower than the applied threshold (min. intensity 7500) was not included in the dataset. Results from the UvA target analysis and the first suspect screening step were used to confirm target OMPs detection in NTA and manually assign ID level 1 when detected with both target and non-target analyses.

2.5. Data interpretation

The average OMP composition (i.e., type and concentration of OMPs) at sampling points I1 and I2 (Fig. 1) was used as the reference for infiltrated water composition in the interpretation of target and non-target results. This accounts for the high variability in OMP concentration in surface and pond water. If the concentration at I1 or I2 was below the limit of quantification (LOQ, Table S4), half of the LOQ was assigned to the non-detected sample for averaging. Historical data on water composition in well I was used to support our observation and previous detection of OMPs (Supplementary Excel, Table E1). Vitens' target data were used to calculate OMP removal efficiencies according to Eq. (1), where $C_{i,inf.water}$ and $C_{i,AW}$ are the concentrations of the OMP (i) in the infiltrated water and the abstraction well, respectively.

Eq. (1). Removal efficiency of an OMP (i).

$$RE_i = \frac{\left(C_{i,\ inf.water} - C_{i,\ AW}\right)}{C_{i,inf.water}} \tag{1}$$

For per- and polyfluoroalkyl substances (PFAS), the relative potency factor (RPF) was used to convert their concentrations to perfluorooctanoic acid (PFOA) equivalents, allowing comparison with environmental regulations (Directorate-General for Environment, 2022b).

2.6. Mobility and persistence data collection

The log D_{ow} is the pH-dependent hydrophobicity proxy that takes OMP speciation into account and was used as a rough estimate of OMP mobility. The log D_{ow} parameter was calculated for each OMP detected via (Vitens) target analysis at pH 7 via ChemAxon using the SMILES of the OMPs detected in the samples (logD Predictor plugin (ChemAxon, n. d.)). ChemAxon was also used to predict the charge state of OMPs detected via target and non-target analyses at pH 7.

To estimate persistence in the MAR system of OMPs detected with target and non-target analysis, Biowin3 scores were obtained from the EPI Suite™ software (US EPA, 2025) using SMILES. Biowin3 estimates the probability of ultimate biodegradation for a given compound, i.e., the time it would take to undergo mineralization under aerobic conditions. In NTA results interpretation, the structure of the candidate with the highest priority (prioritized as explained in Section 2.4.1) was considered for Biowin3 calculations.

3. Results and discussion

3.1. MAR improves water quality for most target OMPs except PFAS

The five WWTPs upstream in the River Berkel affected the quality of the water entering the MAR system, as shown by the detection of OMPs that can be traced back to WWTP origins (Table 1). Particularly, the infiltrated water contained contaminants at relatively high concentrations, like the artificial sweetener sucralose (1.07 μg/L) and the X-ray contrast medium iopamidol (0.72 μg/L), observed also in previous

Table 1
OMPs in the infiltrated water and their removal efficiency (RE) in the abstraction wells within the infiltration area and finished drinking water (mixed with other produced waters), target data. Color code: dark green = RE > 90 %; light green 60 % < RE < 90 %; yellow 40 % < RE < 60 %; red RE < 40 %. Italic font indicates parent compounds. Between parenthesis, the type of OMP is indicated (I = industrial component, P = pesticide, Ph = pharmaceutical). NA = Not measured. * Average concentration of two depths of monitoring well I (half the LOQ was used when one of the two depths was <LOQ).

OMP name	Infiltrated water*	Abstraction well 1		Abstraction well 2		Finished drinking water	
	C (μg/L)	C (μg/L)	RE%	C (μg/L)	RE%	C (μg/L)	RE%
1,3-Diphenylguanidine (I)	0.017	<0.01	< LOQ	<0.01	< LOQ	<0.01	< LOQ
1H-benzotriazole (I)	NA	0.039		NA		NA	
4-methyl-1H-benzotriazole (I)	0.111	0.020	82%	<0.01	< LOQ	<0.01	< LOQ
5-methyl-1H-benzotriazole (I)	0.039	<0.01	< LOQ	<0.01	< LOQ	<0.01	< LOQ
Acesulfame (I)	0.112	0.140	-25%	0.040	64%	0.130	-16%
AMPA (P)	0.075	<0.02	< LOQ	<0.02	< LOQ	<0.02	< LOQ
BAM (P)	0.033	<0,01	< LOQ	0.017	48%	0.018	45%
Bentazon (P)	<0.01	0.014		0.021		0.019	
Carbamazepine (Ph)	0.049	<0.01	< LOQ	<0.01	< LOQ	<0.01	< LOQ
Cyclamate (I)	0.024	0.029	-23%	<0.02	< LOQ	<0.02	< LOQ
DEET (P)	0.009	<0,01	< LOQ	<0.01	< LOQ	<0.01	< LOQ
Diatrizoic acid (Ph)	0.240	<0.05	< LOQ	<0.05	< LOQ	<0.05	< LOQ
Dimethenamide-ESA (P)	0.350	0.190	46%	0.500	-43%	0.160	54%
Dimethenamide-OA (P)	0.150	0.079	47%	0.400	-167%	0.072	52%
Gabapentin (Ph)	0.266	0.135	49%	0.037	86%	0.049	82%
Iohexol (Ph)	0.093	<0.05	< LOQ	<0.05	< LOQ	<0.05	< LOQ
Iomeprol (Ph)	0.019	<0.01	< LOQ	<0.01	< LOQ	<0.01	< LOQ
Iopamidol (Ph)	**0.720**	0.180	75%	0.027	96%	0.130	82%
Irbesartan (Ph)	0.008	<0.01	< LOQ	<0.01	< LOQ	<0.01	< LOQ
Metformin (Ph)	0.008	<0.01	< LOQ	<0.01	< LOQ	<0.01	< LOQ
Metolachlor-ESA (P)	0.465	0.290	38%	0.210	55%	0.170	63%
Metholachlor-OA (P)	0.200	0.130	35%	0.260	-30%	0.120	40%
Phenazone (Ph)	0.008	<0.01	< LOQ	<0.01	< LOQ	<0.01	< LOQ
Primidone (Ph)	0.026	0.025	2%	<0.01	< LOQ	0.010	61%
Sucralose (I)	**1.075**	0.360	67%	0.076	93%	0.100	91%
Sulfamethoxazole (Ph)	0.010	<0.01	< LOQ	<0.01	< LOQ	<0.01	< LOQ
Sulfapyridine (Ph)	0.009	<0.01	< LOQ	<0.01	< LOQ	<0.01	< LOQ
Tetraglyme (I)	0.027	0.024	11%	<0.01	< LOQ	<0.01	< LOQ
Topramezone (P)	0.014	<0.01	< LOQ	<0.01	< LOQ	<0.01	< LOQ
Trans-10,11 dihydro-10,11-dihydroxycarbamazepine *(Ph)*	0.217	0.086	60%	0.013	94%	0.031	86%
Valsartan (Ph)	0.015	<0.01	< LOQ	<0.01	< LOQ	<0.01	< LOQ
Total	**4.385**	**1.741**	**61%**	**1.601**	**63%**	**1.009**	**77%**

monitoring campaign (Supplementary Excel, Table E1). TFA was also detected at relatively high concentrations (2.85 μg/L, Table 2), but still well below the UBA (Umweltbundesamt-German Environmental Agency) proposed limit of 21 μg/L (UBA, 2020). TFA origin is not only related to WWTPs but also to atmospheric deposition or breakdown of fluorinated compounds (Freeling et al., 2020; Joerss et al., 2024), including pesticides that might have been applied to the fields along the waterway.

The target analysis results showed that soil passage in MAR improved the quality of the infiltrated water, providing a better starting point for drinking water production than surface water affected by WWTP effluents. Comparing the infiltrated water with the abstracted water (average of AW1 and AW2), the sum of PFAS concentration decreased from 2.91 to 1.52 μg/L during soil passage, and the total concentration of the other OMPs decreased from 4.38 μg/L to 1.67 μg/L. Excluding PFAS, the total OMP concentration was reduced by 62 %, with 61 % of the OMPs in the infiltrated water showing removal efficiencies higher than 90 %, and 16 OMPs removed to concentrations below the LOQ in AW1. Our results are similar to a recent study on a similar water reuse technology, riverbank filtration (Muschket et al., 2024).

While some OMPs were efficiently removed in the MAR system, others were not, and some even showed increased concentrations in the abstracted water compared to the infiltrated water. Examples are industrial products and pesticide-TPs, previously reported to be persistent in the environment. Artificial sweeteners, like cyclamate, acesulfame, and sucralose, are often considered tracers of anthropogenic contamination (Lange et al., 2012). They showed an average removal of 35 %, but acesulfame and cyclamate concentrations increased in AW1 compared to the infiltrated water. Metolachlor-TPs and dimethenamide-TPs were removed by only 35 to 49 % in the water abstracted from AW1, and exceeded the Groundwater Framework Directive limit of 0.1 μg/L (European Parliament and Council of the European Union, 2006). Their concentrations in AW2 were higher than in the infiltrated water, likely due to formation during soil passage (Krutz et al., 2004; Stamper and Tuovinen, 1998).

PFAS primarily persisted during MAR, with minimal removal observed. Some PFAS compounds showed reduced concentrations in AW1 compared to the infiltrated water, while others had higher concentrations (Table 2), possibly due to temporal variability in PFAS input. According to the hydrological regional model AMIGO (Arcadis et al., 2008; Cirkel, 2017), AW1 receives most of the infiltrated water, explaining the broader PFAS detected in this well. In contrast, many PFAS were not detected in AW2. The observed concentration decrease in AW1 and the absence of many PFAS in AW2 were most likely due to

Table 2
PFAS detected in the infiltrated water and their removal efficiency (RE) based on actual concentrations in ng/L in the abstraction wells within the infiltration area and finished drinking water (mixed with other produced waters), target data. Color code: dark green = RE > 90 %; light green 60 % < RE < 90 %; yellow 40 % < RE < 60 %; red RE < 40 %. The RPF (Relative Potency Factor) is relative to PFOA (expressed in ng PFOA equivalents per liter) (Directorate-General for Environment, 2022a).

PFAS name	RPF	Infiltrated water		Abstraction well 1			Abstraction well 2			Finished drinking water		
		C [ng/L]	C*RPF [ng eq/L]	C [ng/L]	C*RPF [ng eq/L]	RE%	C [ng/L]	C*RPF ng [eq/L]	RE%	C [ng/L]	C*RPF [ng eq/L]	RE%
Heptafluorobutyric acid (HFBA)	NA	NA	-	11.00	-		NA	-		NA	-	
TFA	NA	2885.00	-	1630.00	-	44%	1280.00	-	56%	662.00	-	77%
TFMS	NA	<0.01	-	77.00	-		20.00	-		13.00	-	
Perfluorobutanesulfonamide (FBSA)	NA	0.46	-	0.51	0.51	-11%	<0,3	<0.3	< LOQ	<0,3	<0.3	< LOQ
Perfluorobutanesulfonic acid (PFBS)	0.001	3.4	0.00	2.8	0.00	18%	2	0.002	41%	<2	<2	< LOQ
Perfluorobutanoic acid (PFBA)	0.05	4.85	0.24	4.4	0.22	9%	5.4	0.27	-11%	<2	<2	< LOQ
Perfluoroheptanoic acid (PFHpA)	0.505	2.65	1.34	1.4	0.71	47%	0.79	0.40	70%	0.45	0.23	83%
Perfluorohexanesulfonamide (FHxSA)	NA	<0.2	-	0.21	-		<0.2	-		<0.2	-	
Perfluorohexanesulfonic acid (PFHxS)	0.6	0.975	0.59	0.49	0.29	50%	<0,2	<0.2	< LOQ	<0,2	<0.2	< LOQ
Perfluorohexanoic acid (PFHxA)	0.01	3.6	0.04	2.4	0.02	33%	<1	<1	< LOQ	<1	<1	< LOQ
Perfluorononanoic acid (PFNA)	10	0.305	3.05	0.63	**6.30**	-107%	<0,5	<0.5	< LOQ	<0,5	<0.5	< LOQ
Perfluorooctanesulfonic acid (PFOS)	2	<0.2	<0.2	0.44	0.88		<0.2	<0.2		<0.2	<0.2	
Perfluorooctanoic acid (PFOA)	1	6.95	**6.95**	3	3.00	57%	1.9	1.90	73%	1.4	1.40	80%
Perfluorooctanoic acid (PFOA) branched	1	0.945	0.95	0.63	0.63	33%	0.68	0.68	28%	<0,5	<0.5	< LOQ
Perfluoropentanesulfonic acid (PFPeS)	0.3005	0.345	0.10	0.2	0.06	42%	<0,2	<0.2	< LOQ	<0,2	<0.2	< LOQ
Perfluoropentanoic acid (PFPeA)	0.03	3.45	0.10	2.4	0.07	30%	1.5	0.05	57%	<1	<1	< LOQ
Total		2912.93	**13.36**	1737.51	**12.19**	40%	1312.27	**3.30**	55%	676.85	1.63	77%

dilution with naturally infiltrating groundwater (further discussed in Section 3.4).

In the produced drinking water, the OMP concentrations were lower than in the abstracted water. The concentrations of PFAS and pharmaceuticals were within the proposed limit for the new Water Framework and the Groundwater Directives (Directorate-General for Environment, 2022a). However, pesticide-TPs and the total pesticide concentration (0.56 μg/L) exceeded the limit set by the Drinking Water Framework Directive (European Parliament and Council of the European Union, 2020). Previous studies have shown that the sand filtration technologies applied at the drinking water treatment plant of our study have low efficiencies in OMP removal (Tröger et al., 2020; Wang et al., 2022). Thus, the quality improvement observed from abstracted to produced water was most likely due to the mixing with abstracted water from wells less impacted by the MAR system (Table S2). In fact, AW1 and AW2 contributed only 2.7 % and 3.1 %, respectively, to the finished drinking water (Table S2). Despite their small contribution, the study of AW1 and AW2 was important for evaluating MAR performance, as they were the most impacted by the infiltration. Our results confirm that MAR is an effective preliminary step to drinking water treatment, but it may not be sufficient to comply with environmental regulations without further treatment or mixing with other water sources. Drinking water treatment plants relying solely on simple treatment steps (i.e., sand filtration) could benefit from the implementation of advanced technologies to remove persistent OMPs, like activated carbon filtration, reverse osmosis, or oxidation (Fischer et al., 2019; Piai et al., 2019; Sadia et al., 2024). These advanced treatments would still benefit from MAR as a pretreatment step that reduces the contaminant loads (Tröger et al., 2020).

Overall, MAR improves infiltrated surface water quality and is an important step in the drinking water treatment process. We observed different removal efficiencies for the OMPs detected in the infiltrated water, indicating that, next to dilution, also sorption and (bio)degradation explain the decreased OMP concentrations. These processes are affected by OMPs' intrinsic characteristics, including hydrophobicity and persistence, which we address in Sections 3.3 - 3.5.

3.2. NTA uncovers a wide range of TPs of potential concern

NTA detected a wider range of OMPs, particularly TPs, that were not included in the target analysis, making NTA a powerful tool for a comprehensive water quality assessment. Our results show that NTA detected 50 % more OMPs than target analysis, mainly TPs, which are often overlooked in routine water quality monitoring (Fig. 2, Supplementary Excel Table E3). Despite differences in analytical approaches, OMP detection trends in subsurface water samples align well between target and non-target results. In both cases, fewer OMPs were detected in abstracted water compared to infiltrated water, again confirming that water quality improves during soil passage.

Furthermore, NTA revealed a significantly higher number of OMPs in the finished drinking water than in the water from AW1 (n=+12), and AW2 (n=+11) (Fig. 2). This trend was not observed in the target analysis, where the OMP number in drinking water was lower than in the abstracted water. The additional OMPs detected via NTA in drinking water were mainly pharmaceutical TPs, including TPs from roxithromycin and azithromycin, with substantial confidence level (ID level 3, Supplementary Excel Table E3). This suggests that some TPs may originate from drinking water treatment processes, where the abstracted water was treated and mixed with water from other abstraction fields.

Prediction tools and automated TP screening facilitate treatment of complex environmental data, but despite script parameter optimization for selectivity and sensitivity, the prioritization steps introduced constraints. TPs formed exclusively through abiotic reactions or pathways not covered by the used BioTransformer environmental microbial module were missing from the suspect list. Nonetheless, there is considerable overlap between biotic and abiotic TPs (Helmus et al., 2025), and our primary focus was on biodegradation. Additionally, while NTA detected more OMPs overall, some compounds identified in target analysis were missing in the NTA dataset, particularly some PCs and the entire PFAS category (Fig. 2). Some PFAS were initially detected with NTA but later removed during prioritization due to low ID level (e.g., TFMS), low peak intensity, or detection in blank samples (e.g., TFA). Differences in sample pre-treatment (SPE) and analytical instrumentation between methods may also explain discrepancies in PFAS and PC detection.

Despite the limitations of using NTA, these findings highlight the added value of NTA for water quality assessment. The detection of additional TPs in reused, treated, and produced water helps bridge current gaps in monitoring and highlights the need to include TPs in routine assessments. Our results can support updates to water quality regulations to better reflect the diversity of OMPs in the environment. Overall, NTA complements target analysis by providing a more comprehensive view of water quality.

3.3. OMP charge state gives indications on transport in MAR

The charge state of OMPs influences their distribution between soil and water and therefore affects OMP transport in MAR. Other environmental factors, like pH, transformation processes, hydrology, soil composition, and organic matter content, also contribute to the transport of OMPs (Kodešová et al., 2015; Regnery et al., 2017; Schaffer et al., 2012).

When comparing the influence of OMP charge state and log D_{ow}, charge gave more insights into OMP transport in MAR. The soil-water

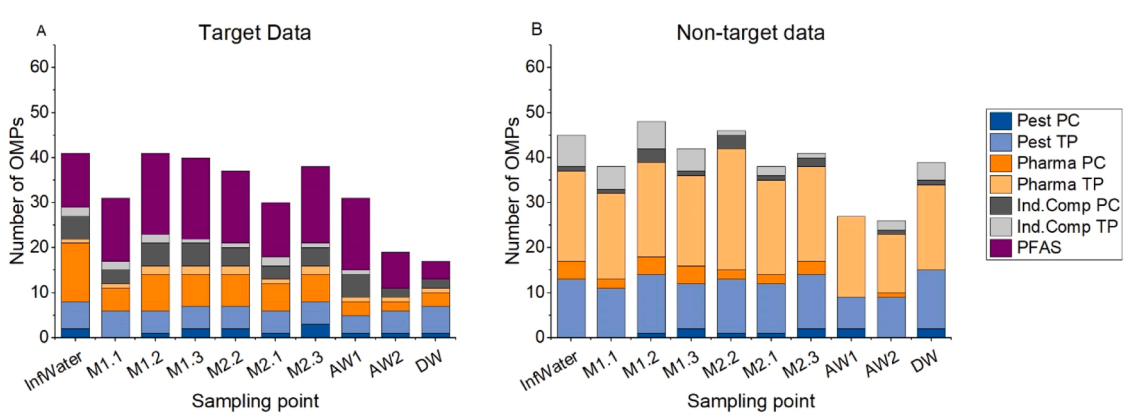


Fig. 2. Comparison of the number of OMPs detected with target analysis and with non-target analysis per sampling point. PC = parent compounds; TP = transformation products; PFAS = per- and polyfluoroalkyl substances; InfWater = infiltrated water; M = monitoring well; AW = abstraction well; DW = drinking water.

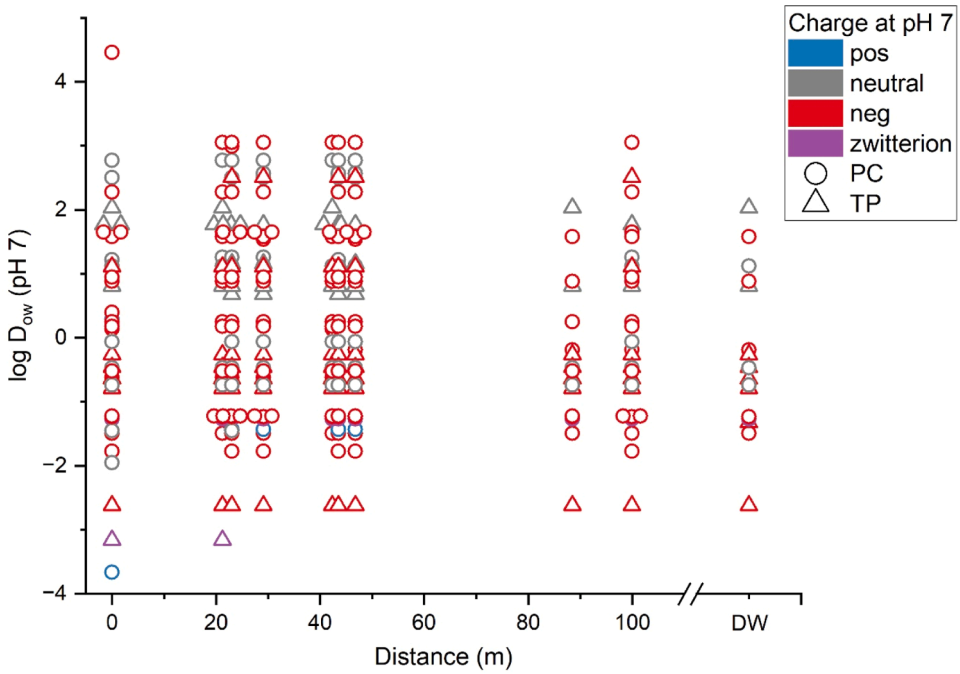
partitioning coefficient normalized to the fraction of organic carbon in soil, K_{oc}, would have been a more suitable parameter than log D_{ow} to describe OMP mobility. All compounds ($n = 23$) for which an experimental log K_{oc} could be retrieved showed values <3, which corresponds to the "mobile" criteria according to the PMT (persistent, mobile, toxic) classification set in the European CLP (classification, labelling, and package) guideline documents (European Parliament and Council of the European Union, 2024). The available experimental log K_{oc} values were well related to the observed OMP mobility, with the median log K_{oc} decreasing from 1.2 to 0.4 from the infiltration location to the abstraction wells, respectively (Figure S2). However, experimental K_{oc} values were missing for 60 % of the OMPs in our study (Figure S3), making the available subset unrepresentative of the full range of OMPs detected in MAR. Thus, the more broadly available log D_{ow} (hydrophobicity proxy) was considered.

Hydrophobicity-based assessment of OMP mobility is widely used to explain OMP transport in the environment (Sigmund et al., 2022). However, log D_{ow} was insufficient to describe the OMP transport in MAR, in line with previous observations (Muschket et al., 2024; Sigmund et al., 2022). Although the results showed significant removal of OMPs with log $D_{ow} > 0$, from 21 OMPs in the infiltrated water to 14 in AW1 and 5 in AW2 (Fig. 3), the log D_{ow} distribution along the traveled distance did not vary substantially (Figure S4). This is likely because OMPs with greater hydrophobicity and lower mobility might have been removed before the water reached the subsurface in the MAR system. For example, these OMPs might have been removed in the WWTP, surface water, or the pond bed.

In contrast, OMP charge state showed correlation with transport in MAR. Neutral and positively charged OMPs were detected mainly within 50 m from the infiltration point (Fig. 3). The zwitterion and positively charged OMPs were detected in low numbers and, except for the zwitterion gabapentin (log D_{ow} −1.27), did not reach either of the abstraction wells, situated at 88 m (AW2) and 99 m (AW1) distance. Only half of the neutral OMPs travelled along the water flow and reached the abstraction wells. On the other hand, most of the negatively charged compounds reached AW1. This group included PFAS, known to be very mobile and persistent (Reemtsma et al., 2016), and OMPs with very low hydrophobicity, such as acesulfame (logD_{ow} −1.49), cyclamate (logD_{ow} −1.77), and dimethenamide-TPs (logD_{ow} −0.79 and −0.64).

A few neutral compounds with log D_{ow} >0 were detected in the abstracted water. Their detection so far from the infiltration pond was not expected based on the observed trend, given the absence of negative charge and relatively high hydrophobicity. These neutral compounds included 1H-benzotriazole (log D_{ow} 1.26), which could have multiple sources, including atmospheric deposition (Zhao et al., 2024). Additionally, some TPs could have formed in the subsurface, like 4-methyl-1-H-benzotriazole (log D_{ow} 1.77) and trans-10, 11-dihydrodihydroxycarbamazepine (log D_{ow} 0.81).

The observed OMP behavior in MAR was likely due to electrostatic interactions, including ion exchange with soil and hydrophobic partitioning. Previous research showed that polar, hydrophylic OMPs can persist in MAR systems with long travel times and stable biogeochemical conditions (Albergamo et al., 2019). Moreover, in our study area, various clay and loam layers are present in the subsurface, though their distribution is uncertain and likely discontinuous (Cirkel, 2017). Given that clay content is positively correlated with soil cation exchange



Distance (m)	0	21.22	23.02	29.10	42.32	43.48	46.79	99.9 (AW1)	88.49 (AW2)	DW
# positive	2	0	1	1	0	1	1	0	0	0
# neutral	13	9	13	12	9	12	11	7	4	5
# negative	23	20	24	24	19	21	23	22	13	11
# zwitterion	2	2	1	1	1	1	1	1	1	1
# logD_{ow} >2	5	4	6	6	3	6	6	3	1	1
# 0 < logD_{ow} < 2	16	12	15	15	12	14	14	11	4	4
# -2 < logD_{ow} < 0	16	13	17	16	13	14	15	15	12	11
# logD_{ow} < -2	3	2	1	1	1	1	1	1	1	1
TOT	**40**	**31**	**39**	**38**	**29**	**35**	**36**	**30**	**18**	**17**

Fig. 3. Graph provides OMPs' log D_{ow} and charge state along the travelled distance from the water infiltration point. The symbol shape indicates if the OMP is a parent compound (PC) or transformation product (TP), and the color indicates the OMP charge state, where pos = positive, and neg = negative charge. The table below indicates the number of OMPs for each category of charge and hydrophobicity (log D_{ow}) along the travelled distance.

capacity and sorption of cationic compounds (Holzmann et al., 2021; Khaledian et al., 2017), and that aromatic compounds can form π-π bonds with the humic and more aromatic fractions of organic matter (Sigmund et al., 2022), the soil composition of the specific MAR system may explain why OMP charge significantly influenced transport.

Since the OMPs detected in the subsurface had a low hydrophobicity and experimental K_{oc} values were largely missing, we could not relate transport behavior to this more robust parameter for soil sorption affinity. However, introducing OMP charge state substantially improved our capacity to better explain their transport in MAR. We believe that this has added value for MAR design in other locations, as the combined study of site-specific soil composition, hydrology, and physico-chemical characteristics of the OMPs in the infiltrated water could help prediction of MAR efficiency and design of field systems.

3.4. PFAS are persistent in MAR systems

PFAS are known to be very persistent chemicals, and this was reflected in our dataset, where concentrations remained largely stable over the travel time points (Fig. 4). In total, 16 PFAS were detected above the LOQ in at least one sampling point along the travel time. Most PFAS detected had a chain length of four or more carbon atoms and, except heptafluorobutyric acid, were detected at ng/L concentrations. Contrarily, shorter chain PFAS, particularly the ultrashort chain TFA, had higher concentrations in the order of μg/L. This is likely because they can be generated from the breakdown of longer-chain PFAS or the (bio)degradation of other OMPs containing a TFA/TFMS moiety (Joerss et al., 2024). PFAS concentrations were generally higher in AW1 than AW2, reflecting the hydrology of the site. According to the AMIGO model, AW1 receives most of the infiltrated water, while AW2 is more influenced by naturally occurring groundwater. This suggests that the PFAS detected in AW1 likely originate from water infiltration in MAR, while the lower concentrations in AW2 are likely due to dilution rather than degradation.

TFA is extremely persistent and mobile, and is thus a chemical of high concern for potential accumulation in the water cycle (Arp et al., 2024). The TFA concentrations detected in our study were higher than the average concentration detected in groundwater in Germany (average 1.36 μg/L (Joerss et al., 2024)) or the Netherlands (average 408 ng/L in groundwater, 971 ng/L in surface water (Schepens et al., 2023)). Concentrations similar to ours were found in the Vechte River, which, like the Berkel, originates from the German region of Münsterland and is heavily affected by pesticide releases (Sturm et al., 2022).

TFA can be seen as a final TP from different sources: pharmaceuticals, pesticides, and volatile TFA-precursors (e.g., refrigerants) (Arp et al., 2024). Due to the volatile precursors and TFA characteristics, rain is one of the major sources of TFA in the environment, together with pesticides (Sturm et al., 2022). The widespread atmospheric deposition and the possible breakdown of TFA-precursors in soil and infiltrating water might explain the higher concentrations detected in our study. Given the observed concentrations in drinking water resources and the documented accumulation potential of TFA in the environment, our findings support the inclusion of TFA in regulatory frameworks.

3.5. TP detection indicates potential for OMP biodegradation in MAR

We used target and non-target analyses to assess TP formation in the MAR system. We assessed the Biowin3 scores and the charge state at pH 7 of the detected OMPs to evaluate their potential biodegradation and mobility. A Biowin3 score below 3 implies mineralization under aerobic conditions would take months or longer, indicating OMP persistence, whereas a threshold of 2.5 distinguishes between "fast" (>2.5) and "slow" (<2.5) biodegradation (US EPA, 2025).

Our results suggest potential biodegradation within MAR, as TPs were detected along the subsurface travel path (Table S7). In our study, 86 feature groups annotated as candidate TPs were detected by NTA, 46 of which were not present in the infiltrated water. To generate the TP suspect list for screening the NTA data, BioTransformer (environmental microbial metabolism option) was used, which predicts biodegradation by environmental microorganisms (Djoumbou-Feunang et al., 2019). This approach, therefore, focused our analysis on TPs most likely formed via biodegradation in MAR. The predicted transformation pathway (Wicker et al., 2016), charge, and the structural information of the TPs are provided in Supplementary Excel Table E3. Most of the 46 TPs detected only in the subsurface were pharmaceutical TPs ($n = 25$), followed by pesticide TPs ($n = 18$) and industrial compound TPs ($n = 3$),

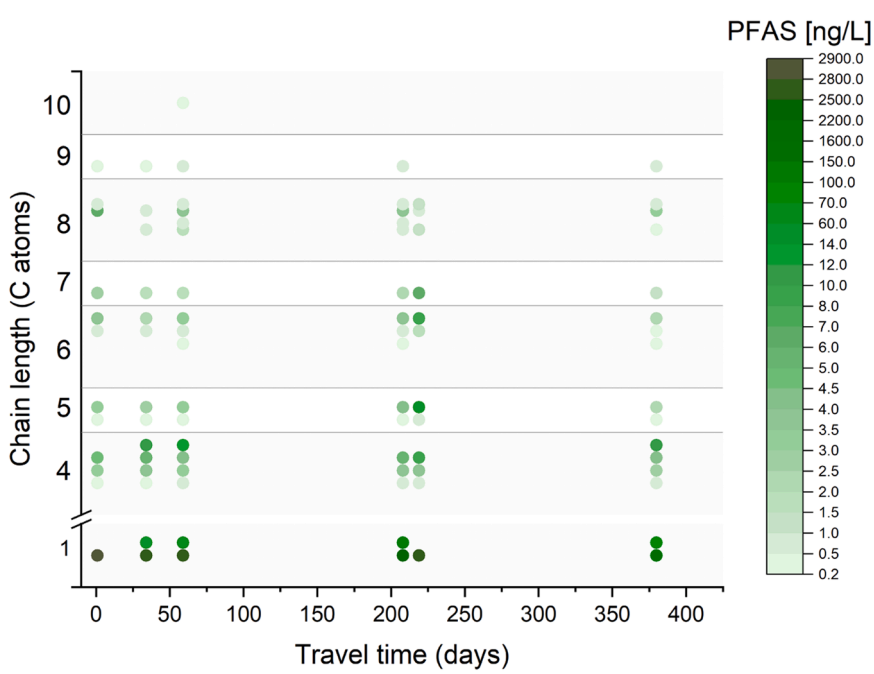


Fig. 4. PFAS concentrations in the MAR system (in ng/L), categorized by carbon-chain length, with the number of carbon atoms (C atoms) on the y-axis (horizontal lines define the group boundaries), and the travel time in days on the x-axis.

indicating that biodegradation and TP formation may occur during soil passage in the MAR system. Previous studies showed biodegradation in MAR (Alidina et al., 2014; Hübner et al., 2022; Maeng et al., 2011) but largely focused on PC removal, with limited insight into TP formation and persistence (Albergamo et al., 2019; Jou-Claus et al., 2024; Oberleitner et al., 2021). For example, X-ray contrast agents showed significant removal in MAR (Maeng et al., 2011), with iopamidol degradation positively correlated with nitrate concentrations (Hamann et al., 2016; Stuyfzand et al., 2007). Our findings align with these results and provide further insights on possible TP formation, as iopamidol-TP candidates were detected under nitrate-reducing conditions (Tables S3, S7, Supplementary Excel Table E3).

Biowin3 scores of the OMPs detected with target analysis were all below 3, suggesting low biodegradation potential (Figure S5). More biodegradable OMPs (Biowin3 >3) were likely removed before entering the MAR system, potentially during wastewater treatment or in surface water, where light and aerobic conditions promote OMP breakdown (Honti et al., 2023; Monteiro and Boxall, 2010). This could be expected as the target list for routine monitoring of the drinking water company likely focused on persistent compounds of potential concern.

We employed Biowin3, developed for aerobic conditions, despite the fact that our MAR system had mainly nitrate-, sulfate-, and iron-reducing conditions, as we found Biowin3 more suitable than Biowin7's focus on methanogenic conditions (Figure S6) (US EPA, 2025). Some discrepancies between the Biowin3 predicted times for biodegradation and our observations remain. For instance, X-ray contrast agents (e.g., iohexol, iomeprol, iopamidol) have Biowin3 scores around 2 (Table S7), yet candidate-TPs were detected within 34 days from infiltration (Supplementary Excel Table E3).

NTA revealed candidate TPs with varying biodegradation potential, as shown by the wider Biowin3 interquartile ranges in Fig. 5, highlighting a greater variability in TP biodegradation potential compared to PCs. Twenty candidate TPs had Biowin3 scores above 3, like phenoxyacetic acid (2,4-dichlorophenoxyacetic acid-TP9), which may be further degraded in the environment. On the other hand, 28 TPs with Biowin3 <2.5 were also detected, including multiple metolachlor-TPs, indicating limited biodegradation potential and possible persistence in the subsurface. Overall, TPs had lower mean and median Biowin3 values than PCs in the subsurface (Fig. 5).

In addition to the biodegradability potential assessment, we also predicted TP charge at pH 7, as electrostatic interactions were shown to be significant for OMP transport in MAR (Section 3.3). Of the 86 TPs, 43 were neutral, 25 negatively charged, 10 positively charged, and 8 zwitterionic. The predominance of neutral and anionic TPs suggests that many may be mobile in MAR, highlighting the need for further monitoring.

These results indicate the possible formation of mobile and persistent TPs during MAR, with potential implications for drinking water quality. While Biowin3 and charge help, toxicological studies would still be needed to complete TP risk assessment. Finally, the discrepancies between Biowin3 predictions and observed degradation suggest the need for models that account for MAR-relevant anaerobic conditions.

4. Conclusions

Our findings highlight MAR as a valuable barrier in drinking water treatment, with >90 % removal efficiency for 61% of the target OMPs. NTA confirmed this trend and revealed TP presence and possible TP formation in the subsurface, suggesting ongoing biodegradation. However, some OMPs persisted and were detected in the drinking water, such as artificial sweeteners, TPs, and PFAS, including TFA. This suggests that MAR could be further optimized by engineering subsurface conditions (Hübner et al., 2022) and complemented by post-MAR technologies for persistent OMPs. NTA was valuable for characterizing the reused, abstracted, and finished water, and it can help draft a more effective target list for routine monitoring.

OMP removal in MAR depends on multiple factors, including intrinsic OMP characteristics and *in-situ* environmental conditions. In

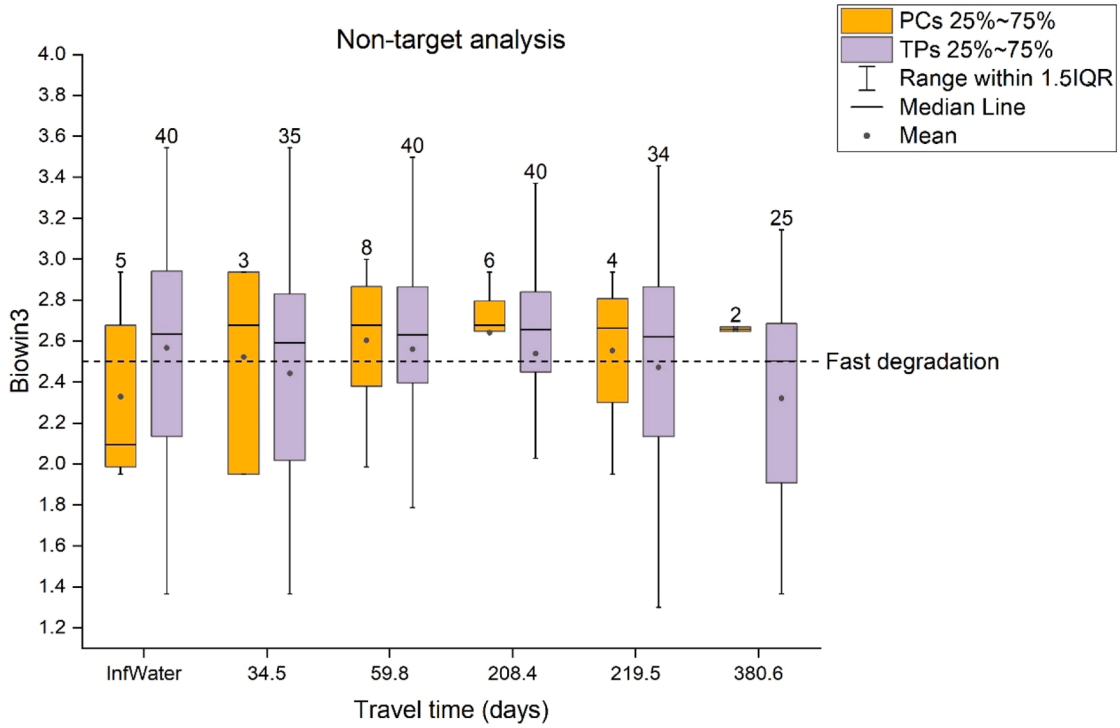


Fig. 5. Trends in Biowin3 scores of the OMPs detected with NTA along the travel time points. PC = parent compound; TP = transformation product; IQR = interquartile range; threshold for fast degradation according to the EPI Suite™ model: Biowin3 = 2.5. The number of PCs and TPs per travel time point is reported on top of each boxplot. The travel time (days) derives from a tracer test performed before this study.

particular, our findings suggest a multidimensional interplay among the intrinsic OMP characteristics of charge state, hydrophobicity, and biodegradation potential. While highly hydrophobic (log D_{ow} >4) and easily biodegradable (Biowin3 >3) OMPs were likely removed in surface water before reaching the MAR system, charge state emerged as a key predictor of transport in MAR for more hydrophilic OMPs. Neutral and positively charged OMPs were mainly removed early in the infiltration pathway, likely due to electrostatic interactions with soil components, while anionic OMPs reached the abstraction wells. The detection of 46 TPs not present in the infiltrated water indicated potential biodegradation in MAR, but also raised concerns about the formation of possibly persistent and mobile TPs. Alongside biodegradable TPs, some persistent TPs, often overlooked in routine monitoring, may form in the subsurface and impact drinking water quality. Further research should address their toxicity and fate under varying redox conditions to improve predictive models and support regulatory inclusion of TPs.

Understanding the combined influence of hydrophobicity, charge state, and biodegradation potential could enhance our knowledge of OMP fate in MAR and inform future MAR system design for better OMP removal. While MAR clearly improves water quality, minimizing the use and production of persistent OMPs remains crucial. Our study emphasizes the need for more comprehensive water quality monitoring that integrates TPs, improved predictive tools, and regulatory adjustments to safeguard drinking water.

CRediT authorship contribution statement

Alessia Ore: Writing – original draft, Visualization, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Gabriel Sigmund:** Writing – review & editing, Data curation, Conceptualization. **Jan Specker:** Writing – review & editing, Methodology, Investigation. **Rick Helmus:** Writing – review & editing, Software, Resources, Methodology, Formal analysis. **Patricia Aguilar-Alarcón:** Writing – review & editing, Validation, Resources, Methodology. **Merijn Schriks:** Writing – review & editing, Resources, Data curation. **Annemarie P. van Wezel:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization. **Nora B. Sutton:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This research was performed within the framework of the research program AquaConnect, funded by the Dutch Research Council (NWO, grant-ID P19–45) and public and private partners of the AquaConnect consortium and coordinated by Wageningen University and Research. We thank Bert Overbeek from Vitens for his support during the sampling campaign and Silvia Ciminale for helping with sample pretreatment.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.watres.2025.124409.

Data availability

Data will be made available on request.

References

Albergamo, V., Helmus, R., de Voogt, P., 2018. Direct injection analysis of polar micropollutants in natural drinking water sources with biphenyl liquid chromatography coupled to high-resolution time-of-flight mass spectrometry. J. Chromatogr. A 1569, 53–61. https://doi.org/10.1016/j.chroma.2018.07.036.

Albergamo, V., Schollée, J.E., Schymanski, E.L., Helmus, R., Timmer, H., Hollender, J., De Voogt, P., 2019. Nontarget screening reveals time trends of polar micropollutants in a riverbank filtration system. Environ. Sci. Technol. 53, 7584–7594. https://doi.org/10.1021/acs.est.9b01750.

Alidina, M., Li, D., Ouf, M., Drewes, J.E., 2014. Role of primary substrate composition and concentration on attenuation of trace organic chemicals in managed aquifer recharge systems. J. Environ. Manage. 144, 58–66. https://doi.org/10.1016/j.jenvman.2014.04.032.

Arcadis, 2008. Rijn En IJssel Water Board, Province of Gelderland, Vitens. AMIGO - Actueel Model Instrument Gelderland Oost.

Arp, H.P.H., Gredelj, A., Glüge, J., Scheringer, M., Cousins, I.T., 2024. The global threat from the irreversible accumulation of trifluoroacetic acid (TFA). Environ. Sci. Technol. 58, 19925–19935. https://doi.org/10.1021/acs.est.4c06189.

ChemAxon, n.d. LogD predictor - chemicalize [WWW Document]. URL https://chemicalize.com (accessed 11.5.24).

Cirkel, G., 2017. Kwantificeren Korte Reistijden Door Middel Van traceronderzoek: Tracerproef Olden Eibergen (BTO Rapport No. BTO 2017.010), Managementsamenvatting. KWR Watercycle Research Institue, Nieuwegein.

Dillon, P., Pavelic, P., Declan, P., Beringen, H., Ward, J., 2009. Managed Aquifer recharge: An introduction (No. 13). National Water Commission, Canberra. Waterline Report Series.

Dillon, P., Stuyfzand, P., Grischek, T., Lluria, M., Pyne, R.D.G., Jain, R.C., Bear, J., Schwarz, J., Wang, W., Fernandez, E., Stefan, C., Pettenati, M., van der Gun, J., Sprenger, C., Massmann, G., Scanlon, B.R., Xanke, J., Jokela, P., Zheng, Y., Rossetto, R., Shamrukh, M., Pavelic, P., Murray, E., Ross, A., Bonilla Valverde, J.P., Palma Nava, A., Ansems, N., Posavec, K., Ha, K., Martin, R., Sapiano, M., 2019. Sixty years of global progress in managed aquifer recharge. Hydrogeol. J. 27, 1–30. https://doi.org/10.1007/s10040-018-1841-z.

Directorate-General for Environment, 2022a. Proposal for a directive amending the Water Framework Directive, the Groundwater Directive and the Environmental Quality Standards Directive.

Directorate-General for Environment, 2022b Annexes to the "Proposal for a Directive of the European parliament and of the Council amending Directive 2000/60/EC establishing a framework for Community action in the field of water policy, Directive 2006/118/EC on the protection of groundwater against pollution and deterioration and Directive 2008/105/EC on environmental quality standards in the field of water policy".

Djoumbou-Feunang, Y., Fiamoncini, J., Gil-de-la-Fuente, A., Greiner, R., Manach, C., Wishart, D.S., 2019. BioTransformer: a comprehensive computational tool for small molecule metabolism prediction and metabolite identification. J. Cheminformat. 11, 2. https://doi.org/10.1186/s13321-018-0324-5.

Drewes, J.E., 2009. Ground water replenishment with recycled water - water quality improvements during managed aquifer recharge. Ground Water 47, 502–505. https://doi.org/10.1111/j.1745-6584.2009.00587_5.x.

Escher, B.I., Fenner, K., 2011. Recent advances in environmental risk assessment of transformation products. Environ. Sci. Technol. 45, 3835–3847. https://doi.org/10.1021/es1030799.

European Parliament, Council of the European Union, 2024. Regulation (EU) 2024/2865 of the European Parliament and of the Council of 23 October 2024 amending Regulation (EC) No 1272/2008 on classification, labelling and packaging of substances and mixtures (Text with EEA relevance), 2024/2865.

European Parliament, Council of the European Union, 2020. Directive (EU) 2020/2184 of the European Parliament and of the Council on the quality of water intended for human consumption, L435/1.

European Parliament, Council of the European Union, 2006. Directive 2006/118/EC on the protection of groundwater against pollution and deterioration. Official J. Eur. Union.

European Parliament, Council of the European Union, 2000. Directive 2000/60/EC establishing a framework for Community action in the field of water policy. Official J. Eur. Union.

Fischer, A., van Wezel, A.P., Hollender, J., Cornelissen, E., Hofman, R., van der Hoek, J. P., 2019. Development and application of relevance and reliability criteria for water treatment removal efficiencies of chemicals of emerging concern. Water Res. 161, 274–287. https://doi.org/10.1016/j.watres.2019.05.088.

Freeling, F., Behringer, D., Heydel, F., Scheurer, M., Ternes, T.A., Nödler, K., 2020. Trifluoroacetate in precipitation: deriving a benchmark data set. Environ. Sci. Technol. 54, 11210–11219. https://doi.org/10.1021/acs.est.0c02910.

Hamann, E., Stuyfzand, P.J., Greskowiak, J., Timmer, H., Massmann, G., 2016. The fate of organic micropollutants during long-term/long-distance river bank filtration. Sci. Total Environ. 545–546, 629–640. https://doi.org/10.1016/j.scitotenv.2015.12.057.

Helmus, R., 2024. patRoon (2.3.3) GitHub repository - workflows for mass-spectrometry based non-target analysis [WWW Document]. https://rickhelmus.github.io/patRoon/index.html (accessed 3.21.24).

Helmus, R., Bagdonaite, I., de Voogt, P., van Bommel, M.R., Schymanski, E.L., van Wezel, A.P., ter Laak, T.L., 2025. Comprehensive mass spectrometry workflows to systematically elucidate transformation processes of organic micropollutants: a case study on the photodegradation of four pharmaceuticals. Environ. Sci. Technol. 59, 3723–3736. https://doi.org/10.1021/acs.est.4c09121.

Holzmann, H., Simeoni, A., Schäffer, A., 2021. Influence of chemical charge on the fate of organic chemicals in sediment particle size fractions. Chemosphere 265, 129105. https://doi.org/10.1016/j.chemosphere.2020.129105.

Honti, M., Zsugyel, M., Seller, C., Fenner, K., 2023. Benchmarking the persistence of active pharmaceutical ingredients in river systems. Environ. Sci. Technol. 57, 14684–14693. https://doi.org/10.1021/acs.est.3c01627.

Hübner, U., Wurzbacher, C., Helbling, D.E., Drewes, J.E., 2022. Engineering of managed aquifer recharge systems to optimize biotransformation of trace organic chemicals. Curr. Opin. Environ. Sci. Health 27, 100343. https://doi.org/10.1016/j.coesh.2022.100343.

Joerss, H., Freeling, F., van Leeuwen, S., Hollender, J., Liu, X., Nödler, K., Wang, Z., Yu, B., Zahn, D., Sigmund, G., 2024. Pesticides can be a substantial source of trifluoroacetate (TFA) to water resources. Environ. Int. 193, 109061. https://doi.org/10.1016/j.envint.2024.109061.

Jou-Claus, S., Rodríguez-Escales, P., Martínez-Landa, L., Diaz-Cruz, M.S., Carrera, J., Sunyer-Caldú, A., Quintana, G., Valhondo, C., 2024. Assessing the fate of benzophenone-type UV filters and transformation products during soil aquifer treatment: the biofilm compartment as bioaccumulator and biodegrader in porous media. Environ. Sci. Technol. 58, 5472–5482. https://doi.org/10.1021/acs.est.3c08465.

Khaledian, Y., Brevik, E.C., Pereira, P., Cerdà, A., Fattah, M.A., Tazikeh, H., 2017. Modeling soil cation exchange capacity in multiple countries. CATENA 158, 194–200. https://doi.org/10.1016/j.catena.2017.07.002.

Kodešová, R., Grabic, R., Kočárek, M., Klement, A., Golovko, O., Fér, M., Nikodem, A., Jakšík, O., 2015. Pharmaceuticals' sorptions relative to properties of thirteen different soils. Sci. Total Environ. 511, 435–443. https://doi.org/10.1016/j.scitotenv.2014.12.088.

Krutz, L.J., Senseman, S.A., McInnes, K.J., Hoffman, D.W., Tierney, D.P., 2004. Adsorption and desorption of metolachlor and metolachlor metabolites in vegetated filter strip and cultivated soil. J. Environ. Qual. 33, 939. https://doi.org/10.2134/jeq2004.0939.

Lange, F.T., Scheurer, M., Brauch, H.-J., 2012. Artificial sweeteners—A recently recognized class of emerging environmental contaminants: a review. Anal. Bioanal. Chem. 403, 2503–2518. https://doi.org/10.1007/s00216-012-5892-z.

Luu, G.T., Freitas, M.A., Lizama-Chamu, I., McCaughey, C.S., Sanchez, L.M., Wang, M., 2022. TIMSCONVERT: a workflow to convert trapped ion mobility data to open data formats. Bioinformatics 38, 4046–4047. https://doi.org/10.1093/bioinformatics/btac419.

Maeng, S.K., Sharma, S.K., Lekkerkerker-Teunissen, K., Amy, G.L., 2011. Occurrence and fate of bulk organic matter and pharmaceutically active compounds in managed aquifer recharge: a review. Water Res 45, 3015–3033. https://doi.org/10.1016/j.watres.2011.02.017.

Monteiro, S.C., Boxall, A.B.A., 2010. Occurrence and fate of human pharmaceuticals in the environment. In: Whitacre, D.M. (Ed.), Reviews of Environmental Contamination and Toxicology. Springer, New York, NY, pp. 53–154. https://doi.org/10.1007/978-1-4419-1157-5_2.

Mumberg, T., Ahrens, L., Wanner, P., 2024. Managed aquifer recharge as a potential pathway of contaminants of emerging concern into groundwater systems – A systematic review. Chemosphere 364, 143030. https://doi.org/10.1016/j.chemosphere.2024.143030.

Muschket, M., Neuwald, I.J., Zahn, D., Seelig, A.H., Kuckelkorn, J., Knepper, T.P., Reemtsma, T., 2024. Fate of persistent and mobile chemicals in the water cycle: from municipal wastewater discharges to river bank filtrate. Water Res 266, 122436. https://doi.org/10.1016/j.watres.2024.122436.

Narain-Ford, D.M., Bartholomeus, R.P., van Wezel, A.P., 2020. Natural purification through soils: risks and opportunities of sewage effluent reuse in sub-surface irrigation. Rev. Environ. Contam. Toxicol. 250, 85–117. https://doi.org/10.1007/398_2020_49.

Oberleitner, D., Schmid, R., Schulz, W., Bergmann, A., Achten, C., 2021. Feature-based molecular networking for identification of organic micropollutants including metabolites by non-target analysis applied to riverbank filtration. Anal. Bioanal. Chem. 413, 5291–5300. https://doi.org/10.1007/s00216-021-03500-7.

Ore, A., Helmus, R., Narain-Ford, D.M., Bartholomeus, R., Sutton, N.B., Van Wezel, A., 2025. Presence of micropollutants and transformation products during subsurface irrigation with treated wastewater assessed by non-target screening analysis. ACS EST Water 2025. https://doi.org/10.1021/acsestwater.4c00930.

Piai, L., Dykstra, J.E., Adishakti, M.G., Blokland, M., Langenhoff, A.A.M., van der Wal, A., 2019. Diffusion of hydrophilic organic micropollutants in granular activated carbon with different pore sizes. Water Res. 162, 518–527. https://doi.org/10.1016/j.watres.2019.06.012.

Reemtsma, T., Berger, U., Arp, H.P.H., Gallard, H., Knepper, T.P., Neumann, M., Quintana, J.B., Voogt, P.de, 2016. Mind the gap: persistent and mobile organic compounds—Water contaminants that slip through. Environ. Sci. Technol. 50, 10308–10315. https://doi.org/10.1021/acs.est.6b03338.

Regnery, J., Gerba, C.P., Dickenson, E.R.V., Drewes, J.E., 2017. The importance of key attenuation factors for microbial and chemical contaminants during managed aquifer recharge: a review. Crit. Rev. Environ. Sci. Technol. 47, 1409–1452. https://doi.org/10.1080/10643389.2017.1369234.

Röst, H.L., Sachsenberg, T., Aiche, S., Bielow, C., Weisser, H., Aicheler, F., Andreotti, S., Ehrlich, H.-C., Gutenbrunner, P., Kenar, E., Liang, X., Nahnsen, S., Nilse, L., Pfeuffer, J., Rosenberger, G., Rurik, M., Schmitt, U., Veit, J., Walzer, M., Wojnar, D., Wolski, W.E., Schilling, O., Choudhary, J.S., Malmström, L., Aebersold, R., Reinert, K., Kohlbacher, O., 2016. OpenMS: a flexible open-source software platform for mass spectrometry data analysis. Nat. Methods 13, 741–748. https://doi.org/10.1038/nmeth.3959.

Sadia, M., Beut, L.B., Pranić, M., Wezel, A.P.van, Laak, T.L.ter, 2024. Sorption of per- and poly-fluoroalkyl substances and their precursors on activated carbon under realistic drinking water conditions: insights into sorbent variability and PFAS structural effects. Heliyon 10. https://doi.org/10.1016/j.heliyon.2024.e25130.

Schaffer, M., Börnick, H., Nödler, K., Licha, T., Worch, E., 2012. Role of cation exchange processes on the sorption influenced transport of cationic β-blockers in aquifer sediments. Water Res. 46, 5472–5482. https://doi.org/10.1016/j.watres.2012.07.013.

Schepens, M., te Biesebeek, J., Hartmann, J., van der Aa, N., Zijlstra, R., Boon, P., 2023. Risk assessment of exposure to PFAS through food and drinking water in the Netherlands. Rijksinstituut voor Volksgezondheid en Milieu RIVM. https://doi.org/10.21945/RIVM-2023-0011.

Schmid, R., Heuckeroth, S., Korf, A., Smirnov, A., Myers, O., Dyrlund, T.S., Bushuiev, R., Murray, K.J., Hoffmann, N., Lu, M., Sarvepalli, A., Zhang, Z., Fleischauer, M., Dührkop, K., Wesner, M., Hoogstra, S.J., Rudt, E., Mokshyna, O., Brungs, C., Ponomarov, K., Mutabdžija, L., Damiani, T., Pudney, C.J., Earll, M., Helmer, P.O., Fallon, T.R., Schulze, T., Rivas-Ubach, A., Bilbao, A., Richter, H., Nothias, L.-F., Wang, M., Orešič, M., Weng, J.-K., Böcker, S., Jeibmann, A., Hayen, H., Karst, U., Dorrestein, P.C., Petras, D., Du, X., Pluskal, T., 2023. Integrative analysis of multimodal mass spectrometry data in MZmine 3. Nat. Biotechnol. 41, 447–449. https://doi.org/10.1038/s41587-023-01690-2.

Sigmund, G., Arp, H.P.H., Aumeier, B.M., Bucheli, T.D., Chefetz, B., Chen, W., Droge, S.T.J., Endo, S., Escher, B.I., Hale, S.E., Hofmann, T., Pignatello, J., Reemtsma, T., Schmidt, T.C., Schönsee, C.D., Scheringer, M., 2022. Sorption and mobility of charged organic compounds: how to confront and overcome limitations in their assessment. Environ. Sci. Technol. 56, 4702–4710. https://doi.org/10.1021/acs.est.2c00570.

Specker, J.C., Praetorius, A., de Baat, M.L., Sutton, N.B., van Wezel, A.P., 2025. Risk characterisation of chemicals of emerging concern in real-life water reuse applications. Environ. Int. 195, 109226. https://doi.org/10.1016/j.envint.2024.109226.

Stamper, D.M., Tuovinen, O.H., 1998. Biodegradation of the acetanilide herbicides alachlor, metolachlor, and propachlor. Crit. Rev. Microbiol. 24, 1–22. https://doi.org/10.1080/10408419891294163.

Sturm, S., Freeling, F., Bauer, F., Vollmer, T., aus der Beek, T., Karges, U., 2022. Trifluoroacetate (TFA): laying the foundations for effective mitigation - spatial analysis of the input pathways into the water cycle (No. (UBA-FB) FB001274/ENG). TEXTE. German Environmental Agency, Dessau-Roßlau, 167/2023.

Stuyfzand, P.J., Segers, W., van Rooijen, N., 2007. Behavior of pharmaceuticals and other emerging pollutants in various artificial recharge systems in the Netherlands, in: ISMAR6. Presented Manag. Aquifer Recharge Sustainab. 231–245.

Tröger, R., Köhler, S.J., Franke, V., Bergstedt, O., Wiberg, K., 2020. A case study of organic micropollutants in a major Swedish water source – Removal efficiency in seven drinking water treatment plants and influence of operational age of granulated active carbon filters. Sci. Total Environ. 706, 135680. https://doi.org/10.1016/j.scitotenv.2019.135680.

UBA, 2020. Trifluoressigsäure (No. ENTWURF_TFA_UQN_Datenblatt). German Environmental Agency.

US EPA, 2025. Estimation Programs Interface Suite™ for Microsoft® Windows.

Wang, J., Poursat, B.A.J., Feng, J., de Ridder, D., Zhang, C., van der Wal, A., Sutton, N.B., 2022. Exploring organic micropollutant biodegradation under dynamic substrate loading in rapid sand filters. Water Res. 221, 118832. https://doi.org/10.1016/j.watres.2022.118832.

Wicker, J., Lorsbach, T., Gütlein, M., Schmid, E., Latino, D., Kramer, S., Fenner, K., 2016. enviPath – The environmental contaminant biotransformation pathway resource. Nucleic Acids Res. 44, D502–D508. https://doi.org/10.1093/nar/gkv1229.

Zahn, D., Arp, H.P.H., Fenner, K., Georgi, A., Hafner, J., Hale, S.E., Hollender, J., Letzel, T., Schymanski, E.L., Sigmund, G., Reemtsma, T., 2024. Should transformation products change the way we manage chemicals? Environ. Sci. Technol. 58, 7710–7718. https://doi.org/10.1021/acs.est.4c00125.

Zhao, M.-L., Fu, J., Ji, X., Zhang, J., He, Z., Yang, G.-P., 2024. Comprehensive analysis of benzothiazoles (BTHs), benzotriazoles (BTRs), and benzotriazole ultraviolet absorbers (BUVs) in the western South China Sea: spatial distributions, migration tendencies and ecotoxicological relevance. Water Res. 266, 122372. https://doi.org/10.1016/j.watres.2024.122372.

Zhou, T., Brunetti, G., Ruud, N., Šimůnek, J., Cui, W., Liao, A., Nasta, P., Gao, J., Levintal, E., García, C.P., Dahlke, H.E., 2025. Simulation of pesticide transport in 70-m-thick soil profiles in response to large water applications. J. Hazard. Mater. 489, 137517. https://doi.org/10.1016/j.jhazmat.2025.137517.